

The Logic Puzzle Walkthrough

This document provides a step-by-step solution to the "Three Knights" logic puzzle.

The Puzzle

Three knights, Sir Reginald, Sir Percival, and Sir Lancelot, are standing in a line. You are told the following:

1. The knight on the left is wearing a red shield.
2. Sir Percival is not in the middle.
3. Sir Lancelot is to the right of the knight with the red shield.

Who is standing in the middle?

Walkthrough

Step 1: Lay out the known facts.

We have three positions: Left, Middle, and Right.

- **Fact 1:** The knight on the Left is the Red Shield Knight.
- **Fact 2:** Sir Percival is not in the Middle.
- **Fact 3:** Sir Lancelot is to the right of the Red Shield Knight.

Step 2: Use the most definitive clues first.

Fact 1 is the most straightforward. We know the knight on the left is wearing a red shield. We don't know which knight this is yet.

Fact 3 is also very powerful. Sir Lancelot is to the right of the Red Shield Knight. Since we know the Red Shield Knight is on the Left, this means Sir Lancelot must be in the Middle or on the Right.

Step 3: Combine the clues to eliminate possibilities.

Let's combine Fact 1 and Fact 3.

- **Left Position:** Red Shield Knight.
- **Middle/Right Positions:** Sir Lancelot could be in either.

Now, let's bring in Fact 2: Sir Percival is not in the middle. This means Sir Percival must be in the Left or Right position.

Step 4: Deduce the final arrangement.

We have the following knowns:

- The Left position is the Red Shield Knight.
- Sir Lancelot is in the Middle or Right position.

- Sir Percival is in the Left or Right position.

Since the Left position is already occupied by the Red Shield Knight, Sir Percival cannot be on the left. This leaves only one option for Sir Percival: he must be on the Right.

Now we know:

- **Right Position:** Sir Percival.

We are left with two knights and two positions: Sir Lancelot and the Middle position.

Sir Lancelot must be to the right of the Red Shield Knight. We've established Sir Percival is on the far right, so Sir Lancelot cannot be there.

This leaves only one option for Sir Lancelot: he must be in the Middle.

The final arrangement is:

- **Left:** Red Shield Knight (by elimination, this must be Sir Reginald)
- **Middle:** Sir Lancelot
- **Right:** Sir Percival

Conclusion

Based on the process of elimination and logical deduction, **Sir Lancelot** is the knight standing in the middle.